

Arthur Prat-Carrabin

Harvard University | Department of Psychology | Computational Cognitive Neuroscience Lab
arthurpc@g.harvard.edu | +1 646 334 6962 | arthurprat.com

- 2024 - **Harvard University**, Cambridge
Postdoctoral Fellow. Department of Psychology, Computational Cognitive Neuroscience Lab. PI: Samuel Gershman.
- 2018 - 2023 **Columbia University**, New York
Associate Research Scholar. Department of Economics, Cognition and Decision Lab. PI: Michael Woodford. (2018-2022: Postdoctoral Research Scholar; spring 2021: Fellow at the Italian Academy for Advanced Studies in America.)

EDUCATION

- 2013 - 2017 **École Normale Supérieure – Physics Department**, Paris
• PhD Student. Advisor: Rava Azeredo da Silveira.
• Thesis: “Bayesian models of human online inference”
- 2005 - 2009 **École Polytechnique**, Palaiseau (Paris)
• French leading “Grande École d’Ingénieur” (scientific school)
• Courses in mathematics, statistics, econometrics, economics, quantum statistical physics, and mechanics
- 2008 - 2009 **ENSAE**, Paris (*École Nationale de Statistique et d’Administration Économique*)
• National statistics school. Courses in statistics, economics, and finance.
- 2002 - 2005 **Lycée Henri IV**, “Classes Préparatoires,” Paris
• Three-year intensive courses in Mathematics, Physics, and Chemistry.

ACADEMIC VISITS & SUMMER SCHOOLS

- April-May 2024 **University of Zurich**, Department of Economics, Zurich, Switzerland
Academic guest, invited by Christian Ruff, Professor of Neuroeconomics and Decision Neuroscience
- July 2018 **Sloan Nomis Summer School on the Cognitive Foundations of Economic Behavior**, Vitznau, Switzerland
- 2013-2016 **Princeton Neuroscience Institute**, Princeton, NJ
Visiting Student Research Collaborator
(Three visits: Nov. 2013-Feb. 2014, Oct.-Dec. 2015, and Oct.-Nov. 2016)
- Aug. 2016 **Champalimaud Center for the Unknown**, Lisbon, Portugal
CAJAL Course in Computational Neuroscience

PUBLICATIONS & MANUSCRIPTS

- 2026 **Prat-Carrabin, A.**, Yamamoto, R., Gershman, S. *Efficient coding makes and breaks Weber's law*. *bioRxiv*. doi.org/10.64898/2026.08.10.744043
- 2026 **Prat-Carrabin, A.**, Gershman, S. *Toward a precise theory of imprecise representations*. Working paper. arthurprat.com/opinion
- 2026 **Prat-Carrabin, A.***, de Hollander, G.* , Bedi, S., Gershman, S., Ruff, C. *Distributed range adaptation in human parietal encoding of numbers*. *bioRxiv*. doi.org/10.1101/2025.09.25.675916
- 2026 **Prat-Carrabin, A.**, Harl, M., Gershman, S. *Fast efficient coding and sensory adaptation in gain-adaptive recurrent networks*. *Nature Communications*. doi.org/10.1038/s41467-026-73032-0
- 2026 **Prat-Carrabin, A.**, Woodford, M. *Endogenous Precision of the Number Sense*. *eLife*. doi.org/10.7554/eLife.101277
- 2025 **Prat-Carrabin, A.**, Gershman, S. *Bayesian estimation yields anti-Weber variability*. *PNAS Nexus*. doi.org/10.1093/pnasnexus/pgaf275
- 2024 **Prat-Carrabin, A.**, Woodford, M. *Imprecise counting of observations in averaging tasks predicts primacy and recency effects*. *bioRxiv*. doi.org/10.1101/2024.09.29.615676
- 2024 **Prat-Carrabin, A.**, Woodford, M. *Imprecise Probabilistic Inference from Sequential Data*. *Psychological Review*. doi.org/10.1037/rev0000469
- 2024 **Prat-Carrabin, A.**, Meyniel, F., Azeredo da Silveira, R. *Resource-rational account of sequential effects in human prediction*. *eLife*. doi.org/10.7554/eLife.81256
- 2022 **Prat-Carrabin, A.**, Woodford, M. *Efficient coding of numbers explains decisions bias and noise*. *Nature Human Behaviour*. nature.com/articles/s41562-022-01352-4
- 2021 **Prat-Carrabin, A.**, Woodford, M. *Bias and variance of the Bayesian-mean decoder*. In M. Ranzato et al., eds., *Advances in Neural Information Processing Systems 34* (NeurIPS 2021). proceedings.neurips.cc/paper/2021
- 2021 **Prat-Carrabin, A.**, Wilson, R., Cohen, J.D., Azeredo da Silveira, R. *Human Inference in Changing Environments with Temporal Structure*. *Psychological Review*. dx.doi.org/10.1037/rev0000276
- 2021 **Prat-Carrabin, A.**, Meyniel, F., Tsodyks, M., Azeredo da Silveira, R. *Biases and Variability from Costly Bayesian Inference*. *Entropy*. 23(5):603. doi.org/10.3390/e23050603

GRANTS, AWARDS, & DISTINCTIONS

- 2024 **NYU SPiNES Finalist**
New York University – Seminars by Postdocs in Neuroscience: Extramural Series

- 2022 **Paper of the Year Award – Society for NeuroEconomics**
Prat-Carrabin, A., Woodford, M. *Efficient coding of numbers explains decisions bias and noise* (2022). Nature Human Behaviour.
- 2021 **Spotlight Presentation – 35th Conference on Neural Information Processing Systems (NeurIPS 2021)**
Prat-Carrabin, A., Woodford, M. *Bias and variance of the Bayesian-mean decoder*.
- Spring 2021 **Fellowship - The Italian Academy for Advanced Studies in America - Columbia University.** “Art, Humanities, and Neuroscience Fellowship” – Appointment as a Postdoctoral Research Scholar at the Italian Academy.
- 2020 **Poster Spotlight – Society for NeuroEconomics Annual Meeting**
Efficient encoding of numbers explains biased judgments.
- 2016 **Competitive Travel Grant - Hebrew University ELSC Annual Retreat**
Edmond & Lily Safra Center for Brain Science, Hebrew University of Jerusalem.
- 2013 **Fondation Pierre-Gilles de Gennes PhD Fellowship**
- 2007 **“Outstanding Leadership” Mention - Ecole Polytechnique**
“Exceptional student in campus and leadership activities”

TALKS

- 2026-07 Cognitive Science Society Annual Conference (Cogsci), Rational Revision Workshop. *“Imprecise beliefs, polarized beliefs”*.
- 2026-02 Warwick Business School, UK. *“A theory of imprecise representations”*
- 2025-12 Department of Neuroscience, American University. *“Flexible Neural Coding for Adaptive Behavior”*
- 2025-10 Laboratory for Computational Vision (PI: Eero Simoncelli), Center for Computational Neuroscience, Flatiron Institute, New York. *“Adaptive efficient coding of magnitudes”*.
- 2025-10 Center for Brain Science, Harvard University. ‘Neurolunch’ seminar. *“Distributed range adaptation in human parietal encoding of numbers”*
- 2025-10 Society for NeuroEconomics, Annual Meeting, Boston. *“Adaptive efficient coding of numerosity in parietal cortex”*
- 2025-07 Zurich Center for Neuroeconomics, University of Zurich, Switzerland. *“Fast efficient coding and sensory adaptation in gain-adaptive recurrent networks”*
- 2025-02 Center for Brain Science, RIKEN, Tokyo. *“Optimal Representations under Neurocognitive Constraints”*
- 2025-01 Department of Psychology, University of California San Diego. *“Optimal Representations under Neurocognitive Constraints”*
- 2024-11 Harvard Medical School Systems Club, Harvard University. *“The Number Sense under Limited Resources”*

- 2024-10 Human and Machine Cognition Lab (PI: Charley Wu), University of Tübingen. *"The Number Sense under Limited Resources"* (online)
- 2024-10 Learning Memory & Decision Lab (PI: Matt Nassar), Brown University. *"The Number Sense under Limited Resources"*
- 2024-09 Cognition, Brain, and Behavior Research Seminar, Harvard University. *"The Number Sense under Limited Resources"*
- 2024-05 Ruff Lab, Zurich Center for Neuroeconomics, University of Zurich, Switzerland. *"Flexible neural coding of numerosity"*, with G. de Hollander.
- 2023-09 Air Force Center of Excellence in the Neuroscience of Decision-Making, Department of Biomedical Engineering, Columbia University, New York. *"Endogenous Imprecision of the Number Sense"*
- 2022-09 Shenhav Lab, Department of Cognitive, Linguistic & Psychological Sciences, Brown University. *"Imprecise Probabilistic Inference from Sequential Data"*
- 2022-09 Zuckerman Institute Postdoctoral Seminars, Columbia University, New York. *"Constrained representations of numerical magnitudes"*
- 2022-01 Laboratory for Computational Vision (PI: Eero Simoncelli), Center for Computational Neuroscience, Flatiron Institute, New York. *"Bias and variance with efficient coding and Bayesian-mean decoding"*
- 2021-12 *"Spotlight presentation"* – NeurIPS 2021 (35th Conference on Neural Information Processing Systems). *"Bias and variance of the Bayesian-mean decoder"*, online.
- 2021-08 Horga lab, Department of Psychiatry, Columbia University, New York. *"Imprecise Probabilistic Inference from Sequential Data"*
- 2021-02 Computational Perception and Cognition Lab (PI: Alan Stocker), UPenn, Philadelphia. *"Encoding-decoding of numbers explains biased judgments"*
- 2021-02 Italian Academy Seminar, Columbia University, New York. *"Encoding-decoding of numbers explains biased judgments"*
- 2020-10 Neuromatch 3.0. *"Efficient encoding of numbers explains biased judgments"*.
- 2020-02 Cognition and Decision Making Joint Lab Meeting, Columbia University, New York. *"Efficient encoding of numbers explains biased judgments"*
- 2019-02 2019 Sloan-Nomis Workshop on the Cognitive Foundations of Economic Behavior, NYU, New York. *"Encoding-decoding of numbers explains biased judgments"*
- 2017-11 Flowers Lab (PI: Pierre-Yves Oudeyer), INRIA (Institut National de Recherche en Informatique et en Automatique), Bordeaux, France. *"Models of human online inference in the presence of temporal structure"*
- 2017-01 2017 ELSC Annual Retreat in Ein Gedi, Hebrew University of Jerusalem, Israel. *"Modulation of inference by the temporal statistics of stimuli"*

- 2016-08 Champalimaud Center for the Unknown, Lisbon, Portugal. “*Robustness and variability of efficient spiking networks*”, with Lueckmann, J.-M. & Gabor L.
- 2016-03 Laboratoire de Physique Statistique (LPS), École Normale Supérieure, Paris. “*Inference in presence of temporal structure in the signal*”
- 2015-07 Institut des Systèmes Intelligents et de Robotique (ISIR), Université Pierre et Marie Curie, Paris. “*Inference of change points with temporal structure*”
- 2014-04 Laboratoire de Neurosciences Cognitives (LNC), École Normale Supérieure, Paris. “*Inference of change-point stimulus with temporal structure*”
- 2014-02 Princeton Neuroscience Institute (PNI), Princeton, NJ. “*Inference of change-point stimulus with temporal structure*”

POSTERS

- 2023-10 **Prat-Carrabin, A.**, Woodford, M. A Bayesian noisy-memory account of recency effects in averaging tasks. *Society for NeuroEconomics Meeting*. Vancouver, BC.
- 2022-11 **Prat-Carrabin, A.**, Woodford, M. Constrained representations of numerical magnitudes. *3rd Workshop on Mental Effort*. Brown University, Providence, RI.
- 2022-10 **Prat-Carrabin, A.**, Woodford, M. Constrained representations of numerical magnitudes. *Society for NeuroEconomics Meeting 2022*. Arlington, VA.
- 2022-08 **Prat-Carrabin, A.**, Woodford, M. Constrained representations of numerical magnitudes. *Conference on Cognitive Computational Neuroscience (CCN 2022)*. San Francisco, CA.
- 2022-07 **Prat-Carrabin, A.**, Woodford, M. Imprecise Probabilistic Inference from Sequential Data. *Cognitive Science Society Annual Conference (Cogsci)*. Toronto.
- 2022-07 **Prat-Carrabin, A.**, Woodford, M. Imprecise Probabilistic Inference from Sequential Data. *Computational Psychiatry Course (CPC++)*. New York.
- 2021-12 **Prat-Carrabin, A.**, Woodford, M. Bias and variance of the Bayesian-mean decoder. *35th Conference on Neural Information Processing Systems (NeurIPS 2021)*. Online.
- 2021-09 **Prat-Carrabin, A.**, Woodford, M. Imprecise Probabilistic Inference from Sequential Data. *Society for NeuroEconomics Meeting 2021*. Online.
- 2020-10 “*Poster Spotlight*” – **Prat-Carrabin, A.**, Woodford, M. Efficient encoding of numbers explains biased judgments. *Society for NeuroEconomics Meeting 2020*.
- 2020-09 **Prat-Carrabin, A.**, Woodford, M. Efficient encoding of numbers explains biased judgments. *Online Bernstein Conference 2020*.
- 2019-05 **Prat-Carrabin, A.**, Ho, B., Woodford, M. Efficient encoding of numbers explains biased judgments. *Zuckerman Institute Mind Brain Behavior Symposium*, Columbia University, New York, USA.
- 2019-02 **Prat-Carrabin, A.**, Ho, B., Woodford, M. Efficient encoding of numbers explains biased judgments. *Computational and Systems Neuroscience (Cosyne)*, Lisbon.

- 2018-10 **Prat-Carrabin, A.,** Ho, B., Woodford, M. Efficient encoding of numbers explains biased judgments. *Society for Neuro-Economics 2018 Annual Meeting*, The Wharton School, University of Pennsylvania, Philadelphia.
- 2016-05 **Prat-Carrabin, A.,** Azeredo da Silveira, R. Modulation of inference by the temporal statistics of stimuli. *Symposium on Biology of Decision Making 2016 (SBDM)*, Institut du Cerveau et de la Moelle Épineière (ICM), Paris.
- 2014-05 **Prat-Carrabin, A.,** Azeredo da Silveira, R. Inference of change-point signals with temporal structure. *Symposium on Biology of Decision Making 2014 (SBDM)*, Institut du Cerveau et de la Moelle Épineière (ICM), Paris.

COMMUNITY SERVICE

- 2021-2025 **Reviewer:**
- MIT Press (book)
 - Nature Communications
 - Proceedings of the National Academy of Sciences (PNAS)
 - NeurIPS 2026
 - NeurIPS 2025 Workshop Proposals
 - PLOS Computational Biology
 - Communications Psychology
 - CogSci 2025
 - Conference on Reinforcement Learning and Decision Making (RLDM)
 - PCI Registered Reports
 - Scientific Reports
 - American Economic Journal: Microeconomics
 - Open Mind: Discoveries in Cognitive Science
 - Conference on Cognitive Computational Neuroscience
- 2022-09 **Teaching Assistant** – *Barcelona Summer School for Advanced Modeling of Behavior*. In charge of a tutorial and of supervising two group projects.
- 2018-2020 **Co-organizer** – *Cognition and Decision “pre-seminar” for PhDs and postdocs*, Columbia University.
- 2007 **A.S.E. humanitarian association**, Huaviña, Chile
Two-month work on the construction of a local product factory.
- 2007 **President of student orientation retreat**, École Polytechnique
Managed a twelve-person team to organize a €130k four-day event to welcome the 500 freshman students to École Polytechnique
- 2006 **Founder of a student association (“Atypix”)**
Organized a forum to meet Ecole Polytechnique alumni with unexpected careers.

OTHER PROFESSIONAL EXPERIENCE

- S1 2015 **Innhotep**, Innovation Consulting, Paris
Consulting analyst in Tech & Innovation. Main assignment: Réseau de Transport d'Électricité (French electricity transmission operator).
- 2011 - 2014 **Whale Street SAS**, Co-Founder
Social media analysis for financial markets. Startup selected by the City of Paris startup accelerator program. Worked on Natural Language Processing, statistics algorithms, and database administration.
- 2009 - 2011 **InfraRed Capital Partner**, Paris (*formerly HSBC Specialist Investment*)
Motorways and High-Speed railways investments. Worked on financial stress tests, risk analysis, traffic prediction models, legal issues and negotiations.
- 2010 - 2011 **Mobile application development**, Android
Developed ClopClop, a mobile application that locates open retailers nearby.
- 2008 - 2009 **Mathematics & Physics Oral Examiner**, Lycée Henri IV, Paris
Examiner for weekly oral exams of students in the new "Classes Préparatoires aux Études Supérieures".
- 2008 **Financial Agency of the Embassy of France**, New York
Study on investment banks: activity and regulation before and after Bear Stearns.
- 2005 - 2006 **Military training in the French Military Police Force**
Midshipman, in the French "Gendarmerie" Polynesian Base.

OTHER SKILLS & INTERESTS

- Languages **French, English**, and **Spanish** (intermediate level).
- Computer skills **Python**, javascript (good level) ; working knowledge of Django, Matlab, C++, PostgreSQL, Objective-C, Java, Ruby, Stata, Mathematica, SAS, and R.
- Other Interests in contemporary dance, literature, running, kiteboarding, and music.